

Model Card

Strategic Asset Management Plan Generator — Model Documentation

SAMP Builder v1 is a web-based, no-installation tool that guides organizations through a structured 9-module workflow to produce a complete, professional Strategic Asset Management Plan. It generates auto-populated narrative text, interactive charts, risk matrices, and financial indicators — producing a print-ready SAMP document aligned with ISO 55000, ISO 31000, and ASTM E917.

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Type	Decision Support & Document Generation Tool	Phone	+971 52 289 9595
Domain	Strategic Asset Management Planning	Language	English (Arabic bilingual — upcoming)
Deployment	Single-file HTML/JS — no server, no installation	License	Proprietary — All rights reserved

■ ISO 55000	■ ISO 31000	■ ASTM E917	■ ISO 55001	■ FHWA LCCA
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1 — Executive Summary

SAMP Builder v1 is a single-file, browser-based document generation tool designed to make Strategic Asset Management Planning accessible, structured, and standardized. Organizations complete a guided 9-module workflow, enter their asset data, and the tool auto-generates a complete, professional SAMP document with narrative text, charts, risk matrices, and financial indicators — ready to present to boards, councils, or regulators.

PROBLEM	SOLUTION	OUTCOME
• SAMP creation costs \$30K-\$150K via consultants	• 9-module guided workflow	• Complete SAMP in 2-3 hours
• 3-6 months delivery time	• Auto-generated professional narrative	• Print-ready professional document
• No Arabic/bilingual option for GCC	• Charts, risk matrix, financial indicators built in	• Audit trail for regulators and boards
• Complex methodology inaccessible to practitioners	• Single HTML file — zero installation	• ISO 55000 / ISO 31000 aligned

2 — Tool Overview & Purpose

SAMP Builder v1 bridges the gap between international asset management standards and practical field application. It does not require specialized software, IT infrastructure, or consultant engagement — any asset manager with data can produce a standards-aligned SAMP in a single session.

Core Design Principles

■ Standards Alignment	Every module maps directly to ISO 55000, ISO 31000, ASTM E917, or FHWA LCCA requirements.
■ Guided Workflow	9 sequential modules ensure no critical section is missed — each gates the next logically.
■ Auto-Generation	Narrative text, financial summaries, and charts are generated automatically from user inputs.
■ Transparency	All calculations are visible, documented, and explainable — no hidden logic.
■ Human Authority	The tool generates a plan — the organization reviews, validates, and adopts it. Final authority remains human.
■ Accessibility	Zero installation, bilingual support, and GCC currency options make it accessible across the region.

3 — Intended Use & Target Users

Asset Manager Any Mode	Produces complete SAMP for organizational planning, budget justification, and board reporting. Primary user.
Technical Consultant Professional	Builds client SAMPs rapidly with structured methodology. Reduces delivery time from months to hours.
Council / Government Officer Basic	Satisfies regulatory and audit requirements for formal asset management documentation at minimum cost.
Engineering Academic Research	Teaching tool for ISO 55000 methodology — demonstrates all 8 SAMP steps in one interactive system.

Out-of-Scope Use Cases

- ✗ Autonomous plan adoption without expert review and organizational approval
- ✗ Real-time CMMS integration or live asset monitoring
- ✗ Legal or regulatory certification as standalone documentation
- ✗ Financial securities or investment portfolio valuation
- ✗ Individual asset-level detailed engineering analysis (use AssetVal Pro v2 for this)

4 — 9-Module Architecture

SAMP Builder v1 follows a sequential 9-module structure, each corresponding to a recognized section of a compliant Strategic Asset Management Plan per ISO 55000. Each module gates the next, ensuring data completeness before plan generation.

Module 1 Organization Setup	Vision, mission, strategic goals, SMART AM objectives, sector selection with auto-presets.
Module 2 Asset Inventory	Asset register, Gross Replacement Cost, Carrying Value, Depreciation, and auto-calculated Lifecycle Expenditure Indicator.
Module 3 Asset Condition	Condition scores (1-10), Remaining Useful Life, data confidence ratings, and action priorities per asset class.
Module 4 Levels of Service	Three-position service table: Current Performance, Desired Optimum, and Agreed Sustainable Position.
Module 5 Future Demand	Demand driver identification (population, technology, climate, regulation, economic) with renewal vs upgrade split.
Module 6 Lifecycle Management	10-year expenditure forecast across Operation, Maintenance, Renewal, Upgrade/New, and Disposal activities.
Module 7 Risk Management	Critical asset identification, ISO 31000 5x5 risk matrix, risk register with treatment plans and residual risk.
Module 8 Financial Summary	Three key financial indicators, funding gap analysis, 10-year financial forecast chart, and funding strategy.
Module 9 Improvement Plan	Structured improvement task register with responsibilities, resources, and timelines.

5 — Methodology & Algorithms

SAMP Builder v1 implements four established methodologies synthesized into one workflow.

ISO 55000 — Asset Management Framework
Governs the overall SAMP structure, AM objectives development, Levels of Service definition, and MCDM criteria. The 9 modules map directly to ISO 55000 clauses.
ASTM E917 — Life Cycle Cost Analysis
Drives the Lifecycle Management module — operation, maintenance, renewal, upgrade, and disposal cost projections over the planning period.
ISO 31000:2018 — Risk Management
Powers Module 7 — the 5x5 Risk Matrix (PoF x CoF), critical asset identification, risk register, treatment plans, and residual risk assessment.
Financial Indicator Framework
Three indicators: Asset Renewal Funding Ratio, Operating Surplus Ratio, and Net Financial Liabilities Ratio — standard local government and infrastructure financial health metrics.

6 — Financial Engine & Indicators

	Formula	Target	Interpretation
Funding Ratio	Renewal Budget / Required Renewal x 100	90-100%	Below 90% = assets deteriorating; above 100% = h
Ratio	(Op Revenue - Op Expenses) / Op Revenue x 100	> 0%	Negative = subsidizing operations from capital — u
Liabilities Ratio	(Total Liabilities - Financial Assets) / Op Revenue x 100	< 100%	High ratio = excessive debt burden relative to reven
Expenditure Indicator	Actual Spend / Lifecycle Cost x 100	90-110%	Below 90 = underfunded; above 110 = above-avera

Funding Gap Calculation

Annual Funding Gap = Total Required Expenditure - Available Annual Budget

A positive gap indicates underfunding — documented and managed through service trade-offs, borrowings, grants, or service level reductions.

Lifecycle Expenditure Indicator — Key Insight

The Lifecycle Expenditure Indicator is averaged over the 10-year planning period, not calculated annually, to smooth lumpy renewal expenditure. A low indicator does not automatically indicate poor management — it must be interpreted alongside the Risk Management module to determine whether underspending is a conscious risk-based trade-off or genuine neglect.

7 — Risk Assessment Framework

Module 7 implements the ISO 31000:2018 risk management framework through three components.

Score	Level	Action Required
1-4	Low	Monitor — normal maintenance cycle
5-9	Medium	Review — enhanced inspection recommended
10-16	High	Intervene — prioritize remediation planning
17-25	Critical	Immediate action — safety or operational risk

Risk Score = PoF x CoF

Critical Asset Register
Assets where failure causes significant service loss, safety incidents, or financial shock. Identified in Module 7 and drive inspection/maintenance prioritization.
Interactive 5x5 Risk Matrix
Clickable matrix generates Risk Score from PoF and CoF selections. Color-coded Low/Medium/High/Critical with auto-classification.
Risk Register
Tabular documentation of each risk: asset, failure mode, PoF, CoF, score, level, treatment plan, residual risk, and budget classification.

8 — Input / Output Specification

Key Inputs

Module	Key Inputs	Type	Required?
1 — Setup	Org name, sector, vision, mission, goals, objectives	Text/Select	Yes
2 — Inventory	GRC, Carrying Value, Depreciation, Actual Spend per class	Decimal	Yes
3 — Condition	Condition score (1-10), RUL, data confidence	Integer/Select	Yes
4 — LoS	Current, desired, agreed service positions per activity	Text	Recommended
5 — Demand	Demand drivers, renewal/upgrade split %	Select/Text	Recommended
6 — Lifecycle	Annual expenditure per activity per asset class (10 yrs)	Decimal	Yes
7 — Risk	Critical assets, PoF, CoF, treatment plans	Integer/Text	Yes
8 — Financial	Revenue, expenses, liabilities, renewal budget	Decimal	Yes
9 — Improvement	Task, responsibility, timeline	Text	Recommended

Outputs

- Complete SAMP document — print-ready, professional format
- Cover page with organization details, standards badges, author info
- Auto-generated executive summary from all module inputs
- Asset inventory table with Lifecycle Expenditure Indicators
- Condition summary table with RUL and data confidence ratings
- Three key financial indicators with status classification
- Funding gap analysis — annual need vs available budget
- Asset replacement values bar chart (by asset class)
- Lifecycle Expenditure Indicator chart (traffic light colors)
- Condition assessment chart (color-coded by score)
- 10-year lifecycle expenditure stacked bar chart with budget line
- Risk distribution doughnut chart
- 10-year financial forecast chart (required vs budget)
- Improvement plan table with tasks, responsibilities, timelines
- Model Card PDF (this document) accessible via in-app button

9 — Smart Validation & Warnings

Module	Condition	Warning
Module 2	Lifecycle Indicator < 50	■ Severely underfunded — asset deterioration risk
Module 2	Lifecycle Indicator > 130	■ Above-average renewal activity — verify data
Module 3	Condition Score >= 8	■ Critical condition — urgent action required
Module 3	Data Confidence = Low	■ Financial forecasts carry high uncertainty
Module 8	Renewal Funding Ratio < 90%	■ Renewal underfunded — assets deteriorating
Module 8	Operating Surplus < 0%	■ Operating deficit — unsustainable position
Module 9	No improvement tasks added	■ SAMP requires documented improvement actions

10 — Model Limitations & Assumptions

■ Data Quality	The plan quality directly reflects input data quality. Estimated GRC, assumed condition scores, or unvalidated lifecycle costs produce unreliable outputs — GIGO applies.
■ Point Estimates	All financial projections use single-point estimates. No probabilistic or Monte Carlo analysis is performed.
■ No Real-Time Data	Manual data entry only. No integration with CMMS, ERP, GIS, or IoT systems in v1.
■ Narrative Accuracy	Auto-generated narrative text is based on input patterns and templates. Expert review before formal adoption is mandatory.
■ Demand Forecasting	Demand projections are user-entered estimates — not connected to population databases, climate models, or economic forecasts.
■ Currency	Multi-currency support is for labeling only. No FX conversion is performed. All inputs must be in the same currency.

11 — Transparency & Explainability

✓ Open Logic	All calculations performed in client-side JavaScript — fully inspectable via browser developer tools. No hidden algorithms.
✓ Formula Documentation	Every financial indicator formula is documented in this Model Card and in contextual help within the tool.
✓ Auto-Narrative Disclosure	Auto-generated text is clearly based on user inputs — no fabricated data or assumed values are inserted.
✓ Audit Trail	The print report captures all inputs, calculations, and outputs — providing a complete record for management review.
✓ Standards Citation	Every module references the specific international standard that governs its methodology.

12 — Human Oversight & Confidence

SAMP Builder v1 produces a draft plan — not an adopted plan. The distinction is critical. Expert review, community consultation, council or board adoption, and ongoing monitoring are all human responsibilities that cannot be delegated to this tool.

Tool Output	Draft SAMP document for expert review
Adoption Authority	Council, board, or organizational leadership — not the tool
Confidence Level	High for structure and format; moderate for auto-narrative; depends on data quality for financials
Mandatory Review	All auto-generated narrative must be reviewed by a qualified asset management professional
Override Protocol	Organizations may modify any auto-generated content before formal adoption

13 — Ethical Considerations

Non-Discrimination	Identical analytical logic applied to all sectors and organization types. No bias toward any asset class, sector, or organization size.
Data Privacy	All data processed locally in the browser. No data transmitted to any server, stored in the cloud, or shared with any third party.
Fair Representation	Auto-generated narrative fairly represents the data provided — no embellishment or minimization of risks or funding gaps.
Scope Integrity	Intended for professional asset management planning. Misuse for regulatory certification without expert review is explicitly out of scope.

14 — Compliance & Standards Reference

Standard	Full Title	Application in Tool
ISO 55000:2014	Asset Management — Overview, Principles & Terminology	Overall SAMP structure, AM objectives, all 9 modules
ISO 55001:2024	Asset Management — Management Systems Requirements	SMART objectives, audit trail, improvement plan
ISO 31000:2018	Risk Management — Guidelines	Module 7 — 5x5 matrix, PoF/CoF, risk register
ASTM E917	Standard Practice for Measuring LCC	Module 6 — lifecycle cost methodology
TRB HWA LCCA	Life-Cycle Cost Analysis in Pavement Design	Transport asset lifecycle planning
ISO 42001:2023	AI Management Systems	Transparency, explainability, human oversight principle

15 — Author & Contact

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This model card was prepared in accordance with global AI documentation best practices including Google Model Cards (Mitchell et al., 2019), Hugging Face model card guidelines, and ISO/IEC 42001:2023. It accompanies SAMP Builder v1 as its official technical and transparency documentation.